

Cloud Infrastructures

02 - Virtualisation with KVM

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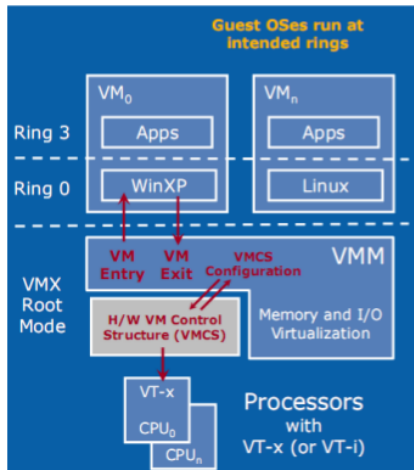
- 1 KVM : Introduction
- 2 CLI-based Management
- 3 Management with Virsh and Libvirt

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Kernel Virtual Machine

Hardware Virtualisation



(Source: Vm Ware)

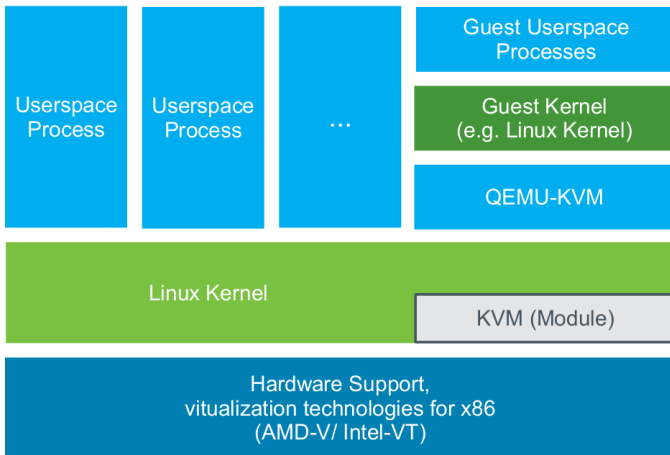
Kernel-based Virtual Machine



- Integrated in the Linux kernel from the 2.6.20 version.
- Aimed at Linux in the x86 platform with virtualisation extensions (Intel VT or AMD-V).
- Comprises a kernel module (`kvm.ko`) to provide the virtualisation infrastructure and a specific module for each processor (`kvm-intel.ko` ou `kvm-amd.ko`).
- It can run guests with Linux or Windows with no modifications.
- Every VM have private, virtualised hardware: network card, disk, graphic card, etc.
- KVM is Open Source software integrated in the kernel Linux. The user-space component is part of QEMU¹.

¹ QEMU is a hosted virtual machine monitor: it emulates the machine's processor through dynamic binary translation and provides a set of different hardware and device models for the machine (Wikipedia).

Architecture



Source: "Virtualization with KVM" training, B1 Systems GmbH

Compatible guests



Solaris, OpenSolaris



Linux



Windows



BSD Unix

Minimum system requirements

- Processor :
 - ▷ An Intel processor with VT-x and Intel 64 extensions for the x86 platform.
 - ▷ An AMD processor with the AMD -V and AMD 64 extensions.
- 6 GB disk.
- 2 GB RAM.
- In general, the VT-x virtualisation option must be enabled in the BIOS. To verify it:
 - ▷ For Intel VT-x: `$ grep --color vmx /proc/cpuinfo`
 - ▷ For AMD V `$ grep --color svm /proc/cpuinfo`

```
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi  
mmx fxsr sse sse2 ss ht tm pbe nx rdtscp lm constant_tsc arch_perfmon pebs bts xtopology  
nonstop_tsc aperfmperf eagerfpu pni pclmulqdq dtes64 monitor ds_cpl vmx est tm2 ssse3 cx16 xtpr  
pdcm pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer xsave avx lahf_lm ida arat epb xsaveopt  
pln pts dtherm tpr_shadow vnmi flexpriority ept vpid
```

Storage support

Storage options for KVM's virtual machines:

- Local files.
- Physical partitions on a disk.
- Physical LUNs (Logical Unit Number) connected locally.
- LVM (Local Volume Manager).
- NFS file systems.
- iSCSI.
- GFS2 cluster file systems.
- Fiber-based LUNs.
- Fibre Channel over Ethernet (FCoE).

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Installation of KVM on Debian/Ubuntu

```
sudo apt-get install qemu-kvm
```

```
sudo apt-get install virt-manager
```

Image formats

■ Qcow2 :

- ▷ QEMU's specific format.
- ▷ Compressed: Variable size (file system independent), the file does not fill all the available space.
- ▷ Advanced functionalities (hot snapshots).

■ Raw :

- ▷ Simple (set of bytes)

Name	Compression	Snapshot	Encryption	Deltas
raw				
qcow2	X	X	X	X
vmdk				X

Creation of a disk image

Disk image:

```
qemu-img create -f qcow2 /data/VMdisk.qcow2 10G
```

```
qemu-img create -f raw /data/VMdisk.img 10G
```

Relative disk image:

- Image creation using another image as a base.
- The new image will exclusively contain the differences from the base image.

```
qemu-img create -b disk.qcow2 -f qcow2 reldisk.qcow2
```

Booting a VM

- Multiple syntaxes.

```
kvm -hda /images/myImage.img \  
-cdrom /isos/SLES-11-DVD-x86_64-GM-DVD1.iso \  
-boot d -m 512 -netdev user,id=user.0,hostfwd=tcp::5555-:22 \  
-device rtl8139,netdev=user.0
```

(man qemu-system-x86_64)

Network

■ Usermode Network Stack (NAT):

- ▷ Default configuration
- ▷ Root rights are not required
- ▷ DHCP, SMB and DNS integrated

■ ▶ TAP device

- ▷ **Public Bridge mode** : the VM is directly accessible from the internet using the host's network card. From a network point of view, the VM will have its own IP address.
- ▷ **Private Bridge mode** : To configure a private network between VMs that is not visible neither from the host nor from the other VMs.

■ Network drivers:

- ▷ rtl8139
- ▷ e1000
- ▷ virtio (paravirtualised)

■ References:

- ▶ <https://help.ubuntu.com/community/KVM/Networking>
- ▶ <http://wiki.qemu.org/Documentation/Networking>
- ▶ Virtual networking: TUN/TAP, MacVLAN, and MacVTap
- ▶ Enabling host-guest networking with KVM, Macvlan and Macvtap

Snapshots

```
qemu-img snapshot -l image.img  
(lists all snapshots)
```

```
qemu-img snapshot -c snapshot image.img  
(creates a snapshot of image.img)
```

```
qemu-img snapshot -a snapshot image.img  
(reverts image.img to the state of the snapshot)
```

```
qemu-img snapshot -d snapshot image.img  
(deletes the snapshot)
```

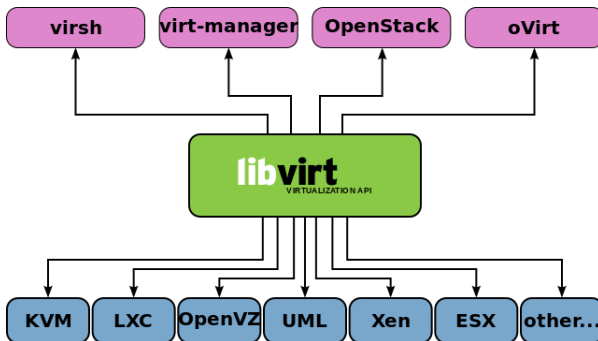
(man qemu-img)

The virtual machine **MUST** be stopped before performing a snapshot!

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Libvirt



- ▀ Libvirt API
- ▀ Reference Manual for libvirt

Information about VMs

■ List VMs:

▷ **virsh list**

▷ **virsh list -all** (also suspended and stopped VMs)

■ Information about nodes:

▷ **virsh nodeinfo**: information about the host node

▷ **virsh dominfo vm_name**: information about the VM

▷ **virsh vcpuinfo vm_name**: information about the VM's CPU

VM creation

```
virt-install \  
--connect qemu:///system \  
--ram=512 \  
--vcpus=2 \  
--name=guest-name \  
--os-type=linux \  
--os-variant=debianwheezy \  
--network=user,model=rtl8139 \  
--disk path=path-to-disk,size=disk-size \  
--cdrom=path-to-install-disk \  
--graphics spice --ram=1024 \  
--keymap=fr
```

Images stored on /var/lib/libvirt/images/

Configuration from XML template files (/etc/libvirt/qemu/), to be copied and modified to create our own VMs

VM Management

■ Boot:

▷ `virsh start vm_name`

■ Stop:

▷ `virsh shutdown vm_name` : soft shutdown

▷ `virsh destroy vm_name` : power down

■ Reboot:

▷ `virsh reboot vm_name`

▷ `virsh shutdown vm_name` et `virsh start vm_name`

■ Suspend:

▷ `virsh suspend vm_name`

Network

- List all the defined networks:
 - ▷ `virsh net-list`
- Get information about networks:
 - ▷ `virsh net-info net_name`
- Activate a network when the host boots:
 - ▷ `virsh net-autostart net_name`
- Start/Stop/Remove network:
 - ▷ `virsh net-start net_name`
 - ▷ `virsh net-destroy net_name`
 - ▷ `virsh net-undefine net_name`

Hibernation

- Save the VM as a file:

- ▷ `virsh save vm_name`

- ▷ `virsh save-image-define save-file.img`

- Restore:

- ▷ `virsh restore save-file.img`

Snapshots

```
virsh snapshot-create-as vm_name snapshot1 "backup1" \  
    --disk-only --atomic  
  
virsh snapshot-list vm_name  
  
virsh snapshot-info vm_name --snapshotname snapshot1
```